Vector of Pos "t"

$$\begin{bmatrix} \sin(\omega t) \\ \cos(\omega t) \end{bmatrix}$$

Vector of Pos "t+δ"

$$\begin{bmatrix} \text{Sin}\,(\omega(t+\delta)) \\ \cos(\omega(t+\delta)) \end{bmatrix}$$

Remember:

$$\sin(a+b) = \sin(a)\cos(b) + \cos(a)\sin(b)$$
$$\cos(a+b) = \cos(a)\cos(b) - \sin(a)\sin(b)$$

$$\sin(\omega(t+\delta)) = \sin(\omega t + \omega\delta) = \sin(\omega t)\cos(\omega\delta) + \cos(\omega t)\sin(\omega\delta)$$

$$\cos(\omega(t+\delta)) = \cos(\omega t + \omega\delta) = \cos(\omega t)\cos(\omega\delta) - \sin(\omega t)\sin(\omega\delta)$$

$$\begin{bmatrix} \text{Sin}\,(\omega(t+\delta)) \\ \cos(\omega(t+\delta)) \end{bmatrix} = \begin{bmatrix} \cos(\omega\delta) & \sin(\omega\delta) \\ -\sin(\omega\delta) & \cos(\omega\delta) \end{bmatrix} \begin{bmatrix} \sin(\omega t) \\ \cos(\omega t) \end{bmatrix}$$

Vector of Pos "t+δ"

Vector of Pos "t"

Independent of Position "t"

From Computer Graphics:

Rotation Matrix of Vector at Pos "t" with angle "ωδ"

e.g "ω", "2ω", "3ω", ...